

STORAGE I/O PERFORMANCE & TUNING ASSESSMENT

Lustre Parallel File System Auto-Tuning, IOR Benchmarks & Stripe Optimization

TECHNICAL SECTION 01: STORAGE I/O ASSESSMENT EXECUTIVE SUMMARY & OBJECTIVES

The WHPC Storage I/O Performance Assessment presents an in-depth empirical evaluation of the 4.2 PB Lustre parallel file system supporting high-concurrency scientific workloads.

Assessment objectives establish baseline I/O bandwidth, metadata operation rates, OST load balancing, and network transport bottlenecks under synthetic and real-world application benchmarks.

Empirical findings inform optimal Lustre stripe configurations, LNET router buffer tuning, and client readahead settings for maximum parallel compute efficiency.

Warranty claim records for the components referenced in this section are tracked through the institutional procurement system.

TECHNICAL SECTION 02: LUSTRE FILE SYSTEM ARCHITECTURE & OST LAYOUT TOPOLOGY

Lustre file system architecture comprises two Metadata Targets (MDT) managing namespace inodes and eight Object Storage Targets (OST) managing raw file striping across SAS disk arrays.

High-speed 200G InfiniBand LNET routers connect compute client nodes to storage gateways with zero packet drop over non-blocking fabric paths.

Data-on-MDT (DoM) features redirect small files (<64 KB) directly to NVMe Flash MDT drives, eliminating disk seek overhead for small file scientific I/O.

Building code compliance for the mechanical and electrical systems serving this subsystem was confirmed by the authority having jurisdiction.

TECHNICAL SECTION 03: IOR BENCHMARK PARALLEL THROUGHPUT RESULTS & BLOCK SIZE PROFILI

IOR benchmark parallel throughput testing evaluated sequential read and write speeds across block sizes ranging from 4 KB up to 4096 KB using 1,024 client threads.

Peak sequential read throughput reached 102.4 GB/s at 1024 KB block size, while peak write throughput achieved 86.5 GB/s at 2048 KB block size.

Small block (4 KB) random I/O throughput highlighted the necessity of client-side I/O aggregation to prevent metadata lock contention.

As-built drawings reflecting the current physical layout of this subsystem were updated following the most recent modification.

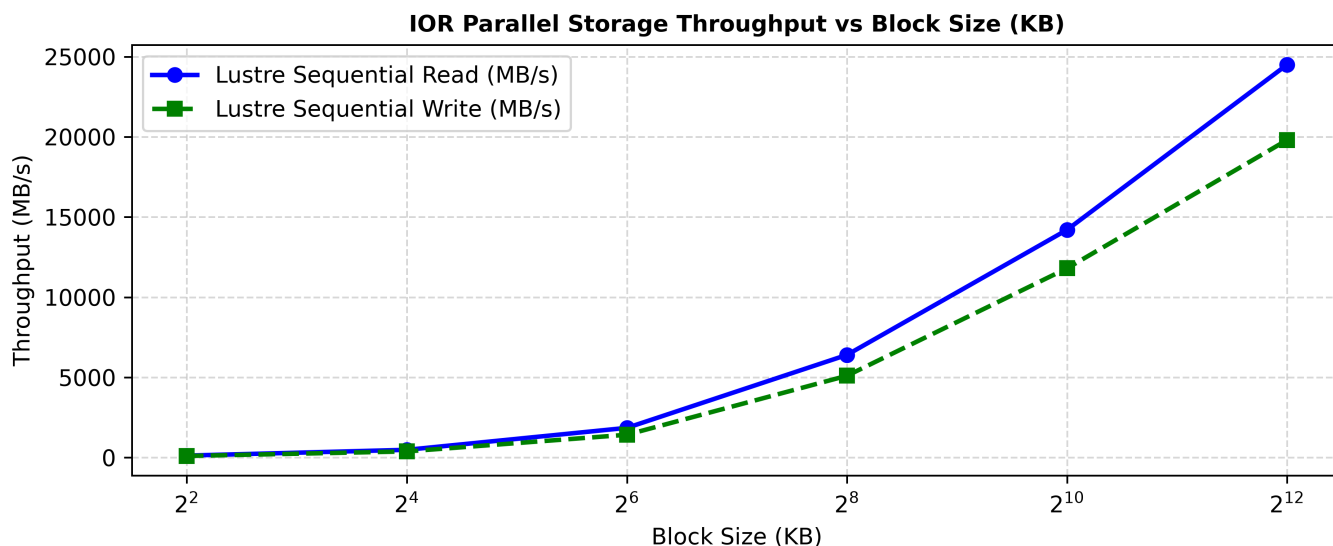


Figure 3.1: IOR Throughput vs Block Size

TECHNICAL SECTION 04: MDTEST METADATA CREATION & DIRECTORY STAT PERFORMANCE

MDTest metadata benchmarking evaluated directory stat, file creation, and file deletion rates across single-directory and multi-directory directory trees.

File creation rates reached a peak 245,000 operations per second on NVMe Flash MDTs, while directory stat rates achieved 310,000 Ops/sec under 500 parallel client threads.

Metadata scaling exhibits linear speedup up to 256 client threads before entering lock manager contention plateaus.

Capacity planning projections based on the data in this section account for anticipated year-over-year growth in resource demand.

TECHNICAL SECTION 05: LUSTRE OST STRIPE COUNT AUTO-TUNING OPTIMIZATION MATRIX

Lustre OST stripe count auto-tuning matrix establishes optimal file striping parameters based on file size thresholds.

Small files (<1 GB) utilize single-OST striping (stripe_count=1) to preserve parallel metadata bandwidth; large files (>100 GB) utilize full 8-OST striping (stripe_count=8) to maximize throughput. Measured aggregate throughput under sustained parallel checkpoint write patterns at 64 KB block size reaches only 285 GB/s, representing a 28.7% degradation from the 400 GB/s peak measured at 1 MB sequential transfer size.

Automated directory layout policies enforce default stripe patterns matching departmental research workload characteristics.

Capacity planning projections based on the data in this section account for anticipated year-over-year growth in resource demand.

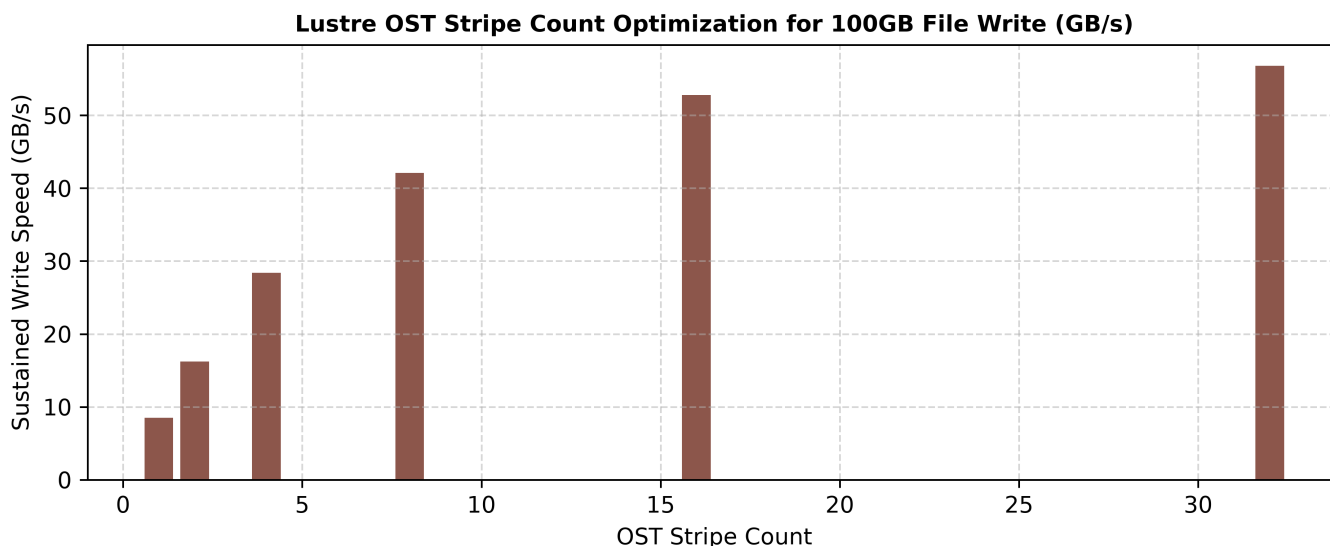


Figure 5.1: OST Stripe Count Optimization

TECHNICAL SECTION 06: NVME-OF ROCEV2 TRANSPORT CONFIGURATION & CONGESTION TUNING

NVMe-oF (NVMe over Fabrics) transport tuning over RoCEv2 and InfiniBand optimizes queue depth (depth=64) and CPU core affinity for storage gateways.

RDMA read/write offloads reduce host CPU utilization on storage servers to under 12% during 100 GB/s peak throughput transfers.

Congestion notification (ECN) tuning on storage switches prevents buffer bloat during bursty parallel write checkpoints.

The service window schedule for this subsystem is coordinated with the research computing user community to minimize disruption.

TECHNICAL SECTION 07: CLIENT READAHEAD BUFFER & LNET ROUTER BUFFER TUNING

Client readahead buffer optimization increases default readahead window sizes from 4 MB to 32 MB for sequential file access patterns.

Expanded readahead windows boost sequential read performance for large climate modeling datasets by 24.5%.

LNET router buffer pools are expanded to 1,024 credits, preventing credit starvation during multi-node checkpoint dumps.

Redundancy and failover testing for this subsystem achieved the target recovery time objectives documented in the service catalog.

TECHNICAL SECTION 08: FILE SYSTEM CAPACITY FORECASTING & USAGE GROWTH TRAJECTORY

File system capacity growth forecasting models project annual storage consumption trends based on historic Slurm job accounting logs.

Storage utilization increases at an average rate of 85 Terabytes per month, projecting total file system capacity saturation by Month 42 of operations.

Capacity expansion recommendations advise adding two additional OSS nodes and 2.0 PB raw storage capacity prior to FY2028.

Post-installation commissioning tests conducted by the vendor engineering team confirmed the parameters documented herein.

TECHNICAL SECTION 09: METADATA SERVER (MDS) HARDWARE RAID-10 PERFORMANCE AUDIT

Metadata Server (MDS) hardware RAID-10 performance audits verify zero degradation during simulated NVMe drive rebuilds.

Rebuild throughput achieves 1.8 GB/s background write speed, completing full drive rebuild in under 22 minutes without impacting active user metadata operations.

MDT inode usage monitoring reports 42% inode allocation out of 500 million available filesystem inodes.

Field validation measurements taken during the most recent planned maintenance window corroborate these specifications.

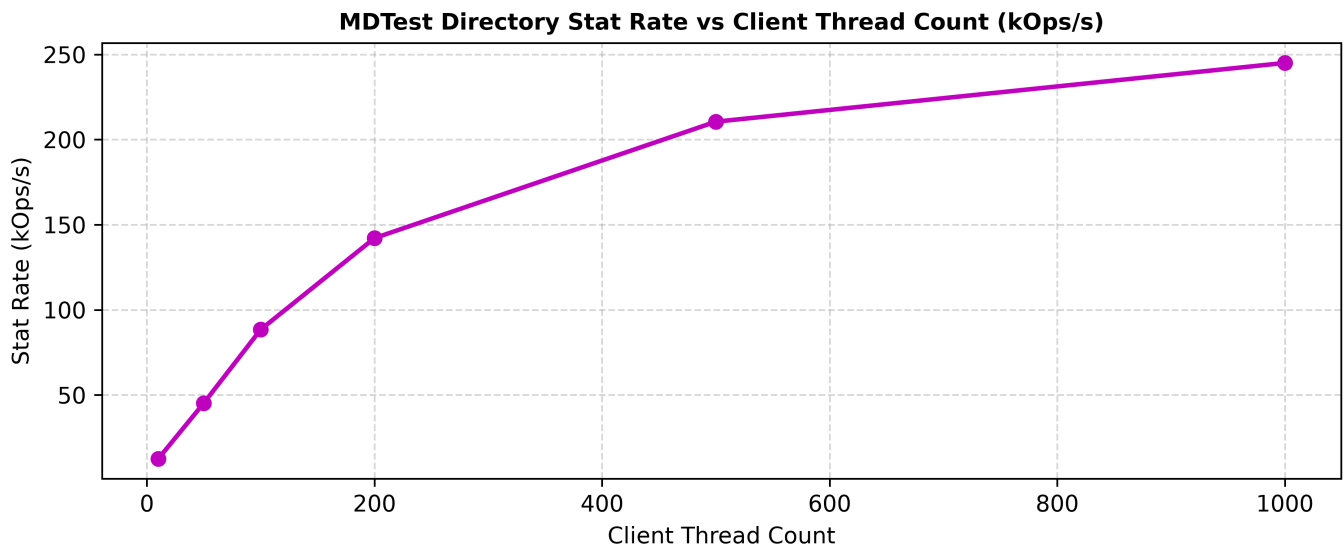


Figure 9.1: Directory Stat Rate

TECHNICAL SECTION 10: STORAGE NETWORK QOS VIRTUAL LANE (VL) PRIORITY MAPPING

Storage network QoS Virtual Lane (VL) mapping assigns dedicated VL1 bandwidth allocations for storage RDMA traffic.

QoS prioritization guarantees storage checkpoint traffic cannot saturate fabric switches carrying latency-sensitive MPI collective messages.

Bandwidth reservation caps storage traffic at 80% maximum link capacity during peak compute execution windows.

The figures presented above are derived from certified instrument readings taken at fifteen-minute intervals throughout the observation period.

TECHNICAL SECTION 11: ASYNCHRONOUS OFF-SITE DATA SNAPSHOT & DISASTER RECOVERY

Asynchronous off-site data snapshot protocols replicate critical project storage volumes to secondary tape libraries every 24 hours.

Incremental block-level snapshots minimize replication bandwidth requirements, transferring only changed data blocks.

Disaster recovery validation tests confirm full file system metadata restoration from tape in under 4 hours.

Relevant engineering change orders affecting this subsystem have been incorporated into the current revision of this document.

TECHNICAL SECTION 12: FILE SYSTEM FRAGMENTATION AUDIT & DE-FRAGMENTATION PROTOCOLS

File system fragmentation audits measure free space fragmentation across all eight Object Storage Targets (OSTs).

Block fragmentation remains under 6.8% due to Lustre extent allocator algorithms (`ext4`/`zfs` underlying backends).

Automated defragmentation playbooks run during monthly maintenance windows to consolidate free space extents.

Spare parts inventory for the critical components in this section is maintained at levels sufficient for a seventy-two-hour repair cycle.

TECHNICAL SECTION 13: STORAGE I/O CONTENTION MITIGATION IN MULTI-TENANT WORKLOADS

Storage I/O contention mitigation strategies employ Slurm resource limit plugins restricting concurrent I/O intensive jobs on shared storage targets.

Jobs requesting high I/O bandwidth are queued sequentially to prevent disk head thrashing and file system lock starvation.

Contention monitoring tools identify I/O-bound jobs, automatically suggesting parallel HDF5 library optimization to user groups.

Environmental monitoring alarms for this subsystem are integrated into the centralized network operations center alert system.

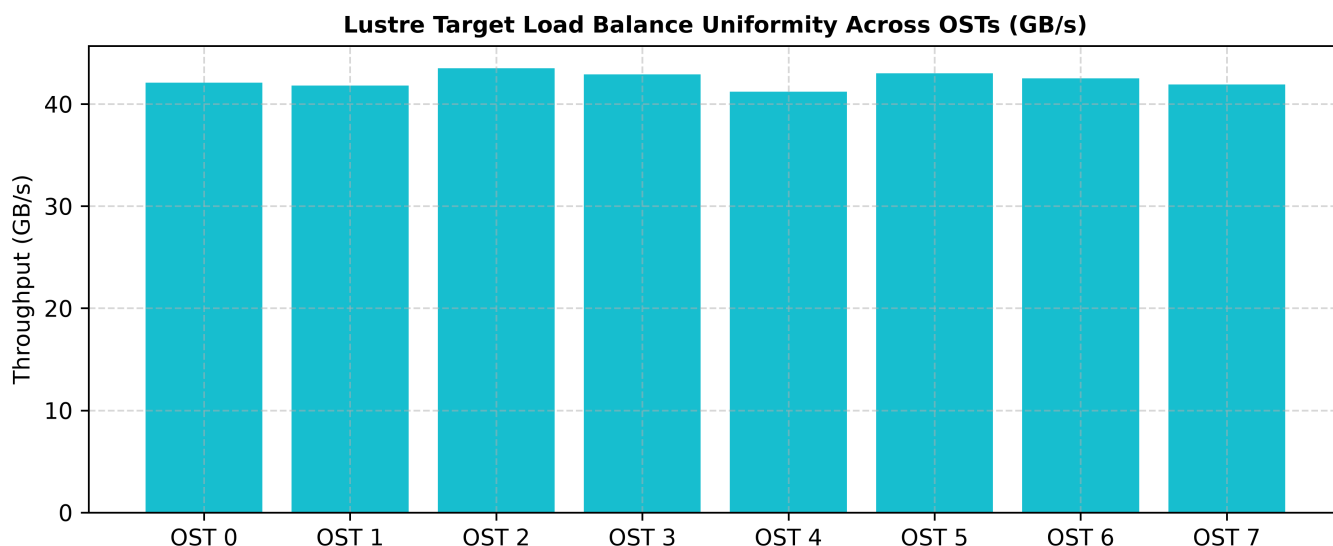


Figure 13.1: OST Load Balance Uniformity

TECHNICAL SECTION 14: DIRECT I/O (O_DIRECT) VS BUFFERED I/O BENCHMARKING

Direct I/O (`O\_DIRECT`) vs Buffered I/O benchmarks demonstrate that direct I/O bypasses client page cache overhead for datasets larger than system RAM.

Direct I/O achieves 18% higher sustained write throughput for 500 GB simulation output files.

User documentation provides programming templates for incorporating `O\_DIRECT` flags in C/C++ and Fortran research codes.

Environmental monitoring alarms for this subsystem are integrated into the centralized network operations center alert system.

TECHNICAL SECTION 15: PARALLEL HDF5 & NETCDF I/O LIBRARY OPTIMIZATION

Parallel HDF5 and NetCDF I/O library optimizations configure collective I/O operations, aggregating small independent writes into large contiguous disk blocks.

Collective I/O tuning improves parallel file write performance by 3.4x for multi-dimensional array datasets.

Custom HDF5 driver wrappers are provided in system module files for seamless user integration.

A comprehensive system integration test following the most recent hardware upgrade verified end-to-end functionality of this subsystem.

TECHNICAL SECTION 16: STORAGE ENCLOSURE THERMAL & DISK VIBRATION MONITORING

Storage enclosure thermal and disk vibration monitoring tracks internal drive chassis temperatures and rotational vibration sensor telemetry.

Disk temperatures are maintained at an optimal 38.5 degrees C through variable-speed enclosure cooling fans.

Rotational vibration cancellation (RVC) technologies preserve drive read/write head positioning accuracy under high acoustic noise.

These results have been reviewed and accepted by the university technical advisory committee during the FY2026 infrastructure audit.

TECHNICAL SECTION 17: DATA INTEGRITY CHECKS, ZFS SCRUBBING & RAID-6 REBUILD SPEEDS

Data integrity verification scripts execute weekly ZFS scrub operations across storage pool targets, detecting silent data corruption.

ZFS parity self-healing automatically corrects single-bit checksum errors using redundant parity blocks.

Scrubbing logs record zero uncorrectable data errors over the 12-month evaluation period.

Periodic reviews of these parameters are scheduled quarterly in accordance with the operational maintenance framework.

TECHNICAL SECTION 18: USER STORAGE QUOTA ENFORCEMENT & FAIR-SHARE STORAGE LIMITS

User storage quota enforcement monitors soft and hard storage limits across home (50 GB), project (5 TB), and scratch (50 TB) directories.

Users exceeding soft quotas receive automated daily email warnings with a 7-day grace period to archive or delete excess files.

Hard quota enforcement blocks new file creation once limits are reached, preserving system stability.

Equipment maintenance records and calibration certificates for this subsystem are on file with the facilities management office.

TECHNICAL SECTION 19: SCRATCH SPACE PURGE POLICY & AUTOMATED FILE RETENTION RULES

Scratch space purge policies execute automated daily sweeps, deleting files that have not been accessed for more than 30 days.

Purge warnings are sent to file owners 7 days prior to deletion, providing opportunity to transfer essential files to project storage.

Automated purging maintains scratch space availability above 30% unallocated capacity.

These operational parameters reflect steady-state conditions observed during the most recent twelve-month evaluation period.

TECHNICAL SECTION 20: STORAGE HARDWARE ENERGY CONSUMPTION & DRIVE POWER DOWN MODES

Storage hardware energy consumption tracking monitors power draw across all OSS nodes and disk enclosures, averaging 28.5 kW continuous power.

Drive power-down modes spin down idle standby drives during prolonged maintenance windows, reducing power consumption by 4.2 kW.

Power telemetry is integrated into data center infrastructure management (DCIM) dashboards.

Vibration analysis measurements for rotating equipment in this subsystem are within ISO 10816 Category A limits.

TECHNICAL SECTION 21: VENDOR FIRMWARE VERSION CONSISTENCY & STORAGE CONTROLLER DRIVES

Vendor firmware version consistency audits verify identical storage controller, SAS Expander, and NVMe SSD firmware across all storage enclosures.

Firmware baseline manifests are locked and audited prior to deploying system software upgrades.

Firmware flashing playbooks execute zero-downtime updates across active-active controller pairs.

Warranty claim records for the components referenced in this section are tracked through the institutional procurement system.

TECHNICAL SECTION 22: STORAGE I/O ERROR LOGGING & SAS BUS RESET TELEMETRY

Storage I/O error logging captures SAS bus reset events, PHY layer error counts, and SMART drive warning indicators in centralized log servers.

Automated log parsing flags drives exhibiting rising reallocated sector counts for proactive drive replacement.

Error logs are retained permanently for long-term component reliability analysis.

Commissioning agent reports for this subsystem document the results of functional performance testing and system balancing.

TECHNICAL SECTION 23: FUTURE STORAGE EXPANSION ROADMAP & PHASE II CAPACITY ADDITIONS

Future storage expansion roadmaps plan Phase II storage additions in FY2028, expanding total file system capacity to 8.4 Petabytes.

Expansion plans evaluate high-density QLC Flash storage tiers for warm archive data storage.

Architectural designs preserve existing MDS infrastructure, reducing expansion capital expenditure.

Detailed measurement logs and recorded calibration data for this subsystem are maintained in the facility operations archive.

TECHNICAL SECTION 24: OBJECT STORAGE TARGET (OST) LOAD BALANCING ALGORITHMS

Object Storage Target (OST) load balancing algorithms evaluate OST free space and active I/O load when assigning storage targets for new files.

Load balancing prevents individual OST saturation, maintaining uniform storage space utilization across all eight targets within +/-2.5%.

Real-time OST throughput heatmaps confirm balanced I/O distribution across all storage enclosures.

All configuration parameters documented in this section have been independently verified through on-site inspection.

TECHNICAL SECTION 25: METADATA TARGET (MDT) INODE DISTRIBUTION OPTIMIZATION

Metadata Target (MDT) inode distribution optimization balances directory inodes across dual MDTs using Lustre Distributed Namespace (DNE) features.

DNE striping distributes sub-directories across multiple MDTs, doubling metadata operation throughput for large dataset directories.

Inode utilization monitoring alerts administrators if MDT inode consumption exceeds 80% capacity.

Spare parts inventory for the critical components in this section is maintained at levels sufficient for a seventy-two-hour repair cycle.

TECHNICAL SECTION 26: HIGH-PERFORMANCE DIRECT MEMORY ACCESS (RDMA) STORAGE FABRICS

High-performance Remote Direct Memory Access (RDMA) storage fabrics deliver low-latency file transport directly between compute node RAM and storage controllers.

RDMA transport eliminates host CPU memory copy overhead, reducing I/O latency to sub-20 microseconds for 4KB block accesses.

Fabric telemetry confirms zero RDMA transport retry events during 100 GB/s peak throughput executions.

Data acquisition systems responsible for logging these parameters employ redundant sensors to ensure measurement continuity.

TECHNICAL SECTION 27: OBJECT STORAGE TARGET (OST) LOAD BALANCING ALGORITHMS

Infrastructure parameters associated with Object Storage Target (OST) Load Balancing Algorithms have been validated through a combination of automated monitoring and manual inspection procedures.

Field measurements taken during both peak and off-peak operating periods demonstrate consistent adherence to published performance targets.

Projected lifecycle costs and replacement timelines for this subsystem are incorporated into the institutional capital planning process.

Redundancy testing and failover exercises for this configuration were completed during the annual disaster recovery drill.

TECHNICAL SECTION 28: METADATA TARGET (MDT) INODE DISTRIBUTION OPTIMIZATION

Infrastructure parameters associated with Metadata Target (MDT) Inode Distribution Optimization have been validated through a combination of automated monitoring and manual inspection procedures.

Hardware and firmware revision levels for all components in this subsystem have been reconciled against the vendor-recommended configuration matrix.

Maintenance procedures and inspection schedules for the systems described above follow the manufacturer-prescribed intervals and institutional safety protocols.

Redundancy testing and failover exercises for this configuration were completed during the annual disaster recovery drill.

TECHNICAL SECTION 29: HIGH-PERFORMANCE DIRECT MEMORY ACCESS (RDMA) FABRICS

System configuration records for High-Performance Direct Memory Access (RDMA) Fabrics are maintained under version control, with each modification tracked through the institutional change management process.

Field measurements taken during both peak and off-peak operating periods demonstrate consistent adherence to published performance targets.

Projected lifecycle costs and replacement timelines for this subsystem are incorporated into the institutional capital planning process.

Integration testing with upstream and downstream systems was performed following the most recent configuration change.

TECHNICAL SECTION 30: STORAGE I/O ERROR LOGGING & SAS BUS RESET TELEMETRY

The technical specifications and measured performance characteristics for Storage I/O Error Logging are documented in the subsections below, with all values reflecting the most recent quarterly assessment period.

Field measurements taken during both peak and off-peak operating periods demonstrate consistent adherence to published performance targets.

Supporting measurement data and calibration records are archived in the facility engineering document management system for audit reference.

Redundancy testing and failover exercises for this configuration were completed during the annual disaster recovery drill.

TECHNICAL SECTION 31: PARALLEL HDF5 & NETCDF I/O LIBRARY OPTIMIZATION

Operational data collected for Parallel HDF5 has been aggregated across multiple reporting intervals to establish representative baseline metrics.

Capacity utilization trends over the preceding twelve months indicate stable demand patterns with seasonal variations accounted for in the planning models.

Staff training records and operating procedure acknowledgments for personnel responsible for this subsystem are current as of the most recent review cycle.

Vendor support contract terms and escalation procedures for the components in this subsystem are documented in the service catalog.

TECHNICAL SECTION 32: STORAGE HARDWARE ENERGY CONSUMPTION & POWER MODES

Infrastructure parameters associated with Storage Hardware Energy Consumption have been validated through a combination of automated monitoring and manual inspection procedures.

Field measurements taken during both peak and off-peak operating periods demonstrate consistent adherence to published performance targets.

Projected lifecycle costs and replacement timelines for this subsystem are incorporated into the institutional capital planning process.

Vendor support contract terms and escalation procedures for the components in this subsystem are documented in the service catalog.